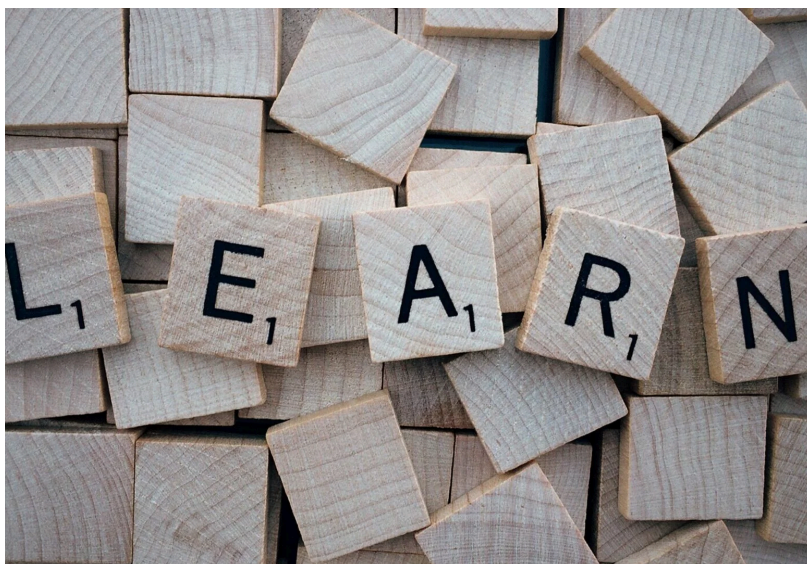


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Effective learning: Twenty rules of formulating knowledge

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Dr Piotr Wozniak, February, 1999 (updated)

[Skip to main content](#) This article will help you overcome one of the greatest difficulties you will face when trying to accelerate learning: formulating knowledge

The speed of learning will depend on the way you formulate the material. The same material can be learned many times faster if well formulated! The difference in speed can be stunning!

The rules are listed in the order of importance. Those listed first are most often violated or bring most benefit if complied with!



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The 20 rules of formulating knowledge in learning

1) Do not learn if you do not understand

Trying to learn things you do not understand may seem like an utmost nonsense. Still, an amazing proportion of students commit the offence of learning without comprehension. Very often they have no other choice! The quality of many textbooks or lecture scripts is deplorable while examination deadlines are unmovable. If you are not a speaker of German, it is still possible to learn a history textbook in German. The book can be crammed word for word. However, the time needed for such "blind learning" is astronomical. Even more important: The value of such knowledge is negligible. If you cram a German book on history, you will still know nothing of history. The German history book example is an extreme. However, the materials you learn may often seem well structured and you may tend to blame yourself for lack of comprehension. Soon you may pollute your learning process with a great deal of useless material that treacherously makes you believe "it will be useful some day".

2) Learn before you memorize

Before you proceed with memorizing individual facts and rules, you need to **build an overall picture of the learned knowledge**. Only when individual pieces fit to build a single coherent structure, will you be able to dramatically reduce the learning time. This is closely related to the problem comprehension mentioned in **Rule 1: Do not learn if you do not understand**. A single separated piece of your picture is like a single German word in the textbook of history. Do not start from memorizing loosely related facts! First read a chapter in your book that puts them together (e.g. *the principles of the internal combustion engine*). Only then proceed with learning using individual questions and answers (e.g. *What moves the pistons in the internal combustion engine?*), etc.

3) Build upon the basics

The picture of the learned whole (as discussed in **Rule 2: Learn before you memorize**) does not have to be complete to the last detail. Just the opposite, the simpler the picture the better. The

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50% of your time repeating just 3-5% of the learned material! Basics are usually easy to retain and take a microscopic proportion of your time. However, each memory lapse on basics can cost you dearly!

4) Stick to the minimum information principle

The material you learn must be formulated in as simple way as it is

- **Simple is easy**

By definition, simple material is easy to remember. This comes from the fact that its simplicity makes it easy for the brain to process it always in the same way. Imagine a labyrinth. When making a repetition of a piece of material, your brain is running through a labyrinth (you can view a neural network as a tangle of paths). While running through the labyrinth, the brain leaves a track on the walls. If it can run in only one unique way, the path is continuous and easy to follow. If there are many combinations, each run may leave a different trace that will interfere with other traces making it difficult to find the exit. The same happens on the cellular level with different synaptic connections being activated at each repetition of complex material

- **Repetitions of simple items are easier to schedule**

I assume you will make repetitions of the learned material using optimum inter-repetition intervals (as in [SuperMemo](#)). If you consider an item that is composed of two sub-items, you will need to make repetitions that are frequent enough to keep the more difficult item in memory. If you split the complex item into sub-items, each can be repeated at its own pace saving your time. Very often, inexperienced students create items that could easily be split into **ten or more** simpler sub-items! Although the number of items increases, the number of repetitions of each item will usually be small enough to greatly outweigh the cost of (1) forgetting the complex item again and again, (2) repeating it in excessively short intervals or (3) actually remembering it only in part!

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Here is a striking example:

Ill-formulated knowledge – Complex and wordy

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(30% by volume) as the ocean. Its density keeps swimmers afloat. Only simple organisms can live in its saline waters

Well-formulated knowledge – Simple and specific

Q: Where is the Dead Sea located?

A: **on the border between Israel and Jordan**

Q: What is the lowest point on the Earth's surface?

A: **The Dead Sea shoreline**

Q: What is the average level on which the Dead Sea is located?

A: **400 meters** (below sea level)

Q: How long is the Dead Sea?

A: **70 km**

Q: How much saltier is the Dead Sea than the oceans?

A: **7 times**

Q: What is the volume content of salt in the Dead Sea?

A: **30%**

Q: Why can the Dead Sea keep swimmers afloat?

A: **due to high salt content**

Q: Why is the Dead Sea called Dead?

A: **because only simple organisms can live in it**

Q: Why only simple organisms can live in the Dead Sea?

A: **because of high salt content**

[Skip to main content](#)

You might want to experiment and try to learn two subjects using the two above approaches and see for yourself what advantage is brought by minimum information principle. This is particularly visible in the long perspective, i.e. **the longer the time you need to remember knowledge, the more you benefit from simplifying your items!**

Note in the example above how short the questions are. Note also that the answers are even shorter! We want a minimum amount of information to be retrieved from memory in a single repetition! **We want answer to be as short as imaginably possible!**

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remedied by adding more questions or making the present ones more precise.

You will also lose the ability to fluently recite the description of the Dead Sea when called up to the blackboard by your teachers. I bet, however, that shining in front of the class is not your ultimate goal in learning. To see how to cope with recitations and poems, read further (section devoted to enumerations)

5) Cloze deletion is easy and effective

Cloze deletion is a sentence with its parts missing and replaced by three dots. *Cloze deletion exercise* is an exercise that uses cloze deletion to ask the student to fill in the gaps marked with the three dots. For example, *Bill ...[name] was the second US president to go through impeachment.* If you are a beginner and if you find it difficult to stick to the minimum information principle, use cloze deletion! If you are an advanced user, you will also like cloze deletion. It is a quick and effective method of converting textbook knowledge into knowledge that can be subject to learning based on spaced repetition. Cloze deletion makes the core of the fast reading and learning technique called *incremental reading*.

III-formulated knowledge – Complex and wordy

Q: What was the history of the Kaleida company?

A: Kaleida, founded to the tune of \$40 million by Apple Computer and IBM in 1991. Hyped as a red-hot startup, Kaleida's mission was to create a multimedia programming language. It finally produced one, called Script X. But it took three years. Meanwhile, companies such as Macromedia and Asymetrix had snapped up all the business. Kaleida closed in 1995.

Well-formulated knowledge – Simple cloze deletion

Q: Kaleida was funded to the tune of ...(amount) by Apple Computer and IBM in 1991

A: \$40 million

Q: Kaleida was funded to the tune of \$40 million by ...(companies) in 1991

A: Apple and IBM

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A: Kaleida's

Q: Kaleida's mission was to create a ... It finally produced one, called Script X. But it took three years

A: multimedia programming language

Q: Kaleida's mission was to create a multimedia programming language. It finally produced one, called ... But it took three years

A: Script X

Q: Kaleida's mission was to create a multimedia programming language. It finally produced one, called Script X. But it took ...(time)

A: three years

Q: Kaleida's mission was to create a multimedia programming language: Script X. But it took three years. Meanwhile, companies such as ... had snapped up all the business

A: Macromedia/Asymetrix

Q: Kaleida's mission was to create Script X. But it took three years. Meanwhile, companies such as Macromedia and Asymetrix had snapped up all the business. Kaleida closed in ...(year)

A: 1995

Optional: SuperMemo

Recipe:

SuperMemo 2002

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Creating *cloze deletions* in new SuperMemos: select the keyword that is to be replaced with tree dots and press *Alt+Z*

SuperMemo 2000

Generating a cloze deletions from texts placed in the clipboard in SuperMemo 2000:

1. Press *Ctrl+Alt+N* to paste the text to SuperMemo

SuperMemo 98/99

Cloze deletions in SuperMemo 98/99:

1. Press *Ctrl+A* to add a standard question-and-answer item
2. Paste the text into the question field. This will create the outline of your items
3. Press *Ctrl+Alt+U* to **Duplicate** the element
4. Select the part that is to be

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	2. Select the part that is to be replaced with three dots 3. Right-click to open the component menu and select Reading : Remember cloze (or click one of cloze icons on the reading toolbar)	replaced with three dots 5. Cut the selection to the clipboard (e.g. with <i>Shift+Del</i>) 6. Type in three dots (optionally, add the explanation in parentheses as in above examples) 7. Press <i>Ctrl+T</i> to save the question field and move to the answer field 8. Paste the text cut in Step 5 (e.g. with <i>Shift+Ins</i> or <i>Ctrl+V</i>). Your first item is ready 9. Press <i>PgUp</i> to go back to the outline item created in Step 2 10. Goto Step 3 and continue adding new items
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6) Use imagery

Visual cortex is that part of the brain in which visual stimuli are interpreted. It has been very well developed in the course of evolution and that is why we say *one picture is worth a thousand words*. Indeed if you look at the number of details kept in a picture and the easiness with which your memory can retain them, you will notice that our verbal processing power is greatly inferior as compared with the visual processing power. The same refers to memory. A graphic representation of information is usually far less volatile. Usually it takes much less time to formulate a simple question-and-answer pair than to find or produce a neat graphic image. This is why you will probably always have to weigh up cost and profits in using graphics in your learning material. Well-employed images will greatly reduce your learning time in areas such as anatomy, geography, geometry, chemistry, history, and many more. The power of imagery explains why the concept of Tony Buzan's mind maps is so popular. A mind map is an abstract picture in which connections between its components reflect the logical connections between individual concepts.

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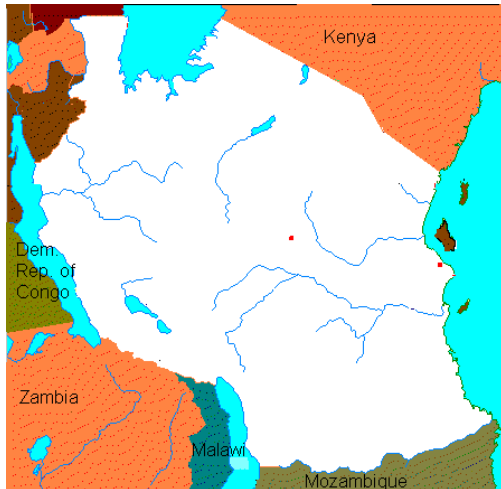


Log in



Well-formulated knowledge – Simple cloze deletion

Q: What African country is marked white on the map?



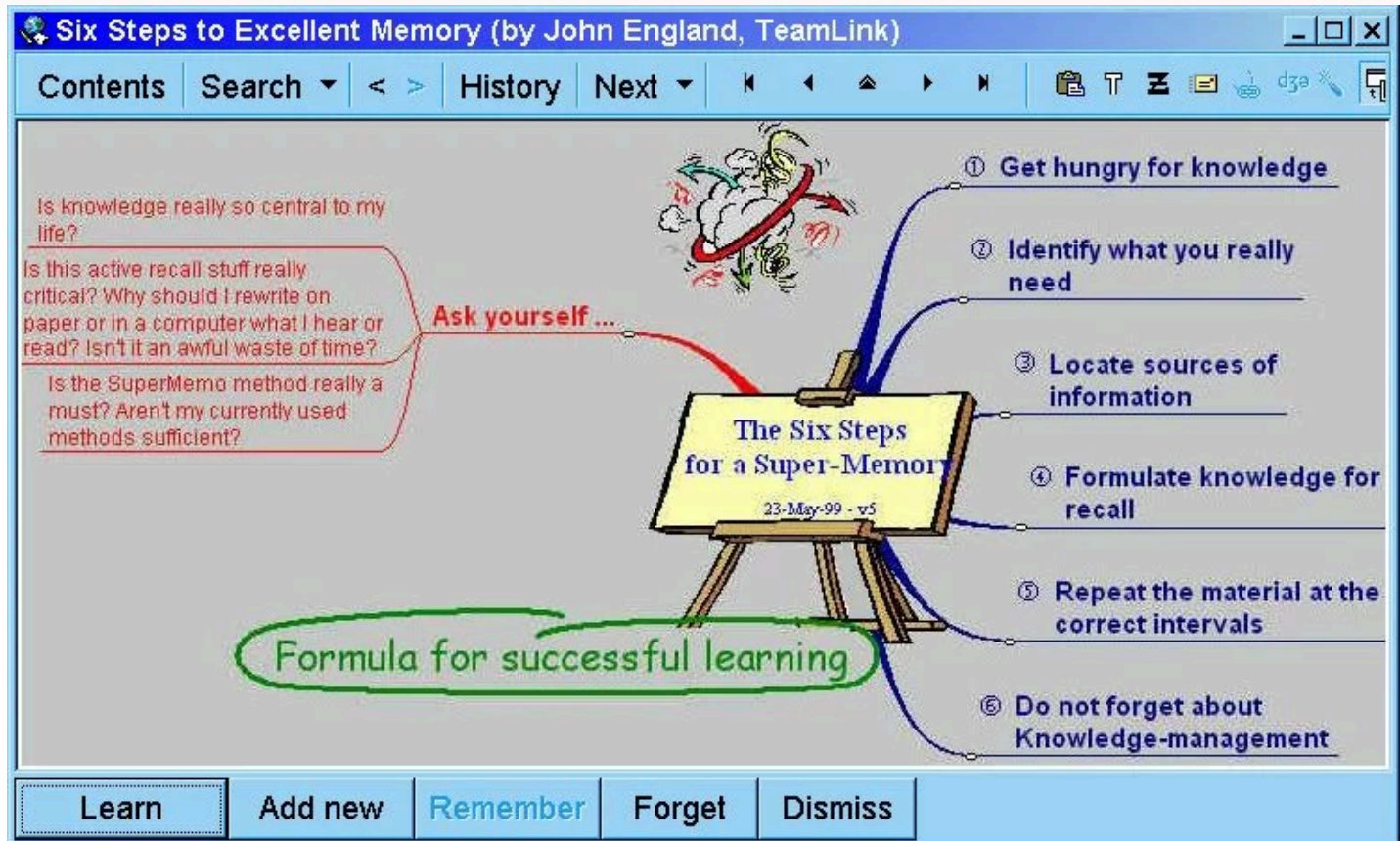
A: Tanzania

7) Use mnemonic techniques

Mnemonic techniques are various techniques that make remembering easier. They are often amazingly effective. For most students, a picture of a 10-year-old memorizing a sequence of 50 playing cards verges on discovering a young genius. It is very surprising then to find out how easy it is to learn the techniques that make it possible with a dose of training. These techniques are available to everyone and do not require any special skills! Before you start believing that mastering such techniques will provide you with an eternal solution to the problem of forgetting, be warned that the true bottleneck towards long-lasting and useful memories is not in quickly memorizing knowledge! This is indeed the easier part. The bottleneck lies in retaining memories for months, years or for lifetime! To accomplish the latter you will need **SuperMemo** and the compliance with the 20 rules presented herein. There have been dozens of books written about mnemonic techniques. Probably those written by Tony Buzan are most popular and respected. You can search the web for keywords such as: *mind maps*, *peg lists*, *mnemonic techniques*, etc. Experience shows that with a dose of training you will need to consciously apply mnemonic

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(Six Steps mind map generated in [Mind Manager 3.5](#), imported to SuperMemo 2004, courtesy of John England, TeamLink Australia)

8) Graphic deletion is as good as cloze deletion

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Graphic deletion works like cloze deletion but instead of a missing phrase it uses a missing image component. For example, when learning anatomy, you might present a complex illustration. Only a small part of it would be missing. The student's job is to name the missing area. The same illustration can be used to formulate 10-20 items! Each item can ask about a specific subcomponent of the image. Graphic deletion works great in learning geography! Exemplary graphic deletion:

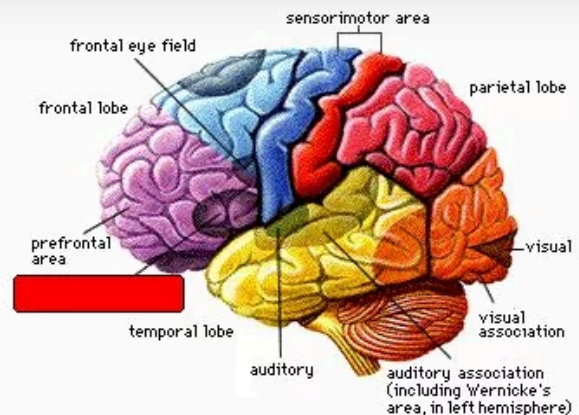
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pointed to by the red rectangle?

Broca's area



Learn

Add new

Remember

Forget

Dismiss

SuperMemo 2000/2002

This is how you can quickly generate graphic deletion using a picture from the clipboard:

1. Press *Shift+Ins* to paste the picture to SuperMemo

2. Press *Ctrl+Shift+M* and choose *Occlusion* template to apply graphic deletion template

3. SuperMemo 2000 only:
Choose *Ctrl+Shift+F2* to impose and detach the *Occlusion* template

4. Fill out the fields and place the

SuperMemo 99

In SuperMemo 99 you will need a few more steps:

1. Create an item containing the following components:

– question text: What is the name of the area covered with the red rectangle?

– empty answer text (click Answer on the component menu)

– your illustration (use Import file on the image component menu)

– red rectangle component (choose red color with Color on the rectangle component menu)

2. Choose Duplicate on the element menu (e.g. by pressing *Ctrl+Alt+U*)

3. *Ctrl+click* the rectangle component twice to place it in the dragging mode

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created in Step 1

7. Go to Step 2 to add generate more graphic deletions

Note that you could also paint covering rectangles or circles on the original image but this would greatly increase the size of your collection. The above method makes sure that you reuse the same image many times in all items of the same template. For example, the collection Brain Anatomy available from >SuperMemo Library and on SuperMemo MegaMix CD-ROM uses the above technique

A more detailed recipe for creating occlusion tests is presented in: Flow of knowledge

9) Avoid sets

A **set** is a collection of objects. For example, a set of fruits might be an apple, a pear and a peach. A classic example of an item that is difficult to learn is an item that asks for the list of the members of a set. For example: *What countries belong to the European Union?* You should avoid such items whenever possible due to the high cost of retaining memories based on sets. If sets are absolutely necessary, you should always try to convert them into enumerations. Enumerations are ordered lists of members (for example, the alphabetical list of the members of the EU). Enumerations are also hard to remember and should be avoided. However, the great advantage of enumerations over sets is that they are ordered and they force the brain to list them always in the same order. An ordered list of countries contains more information than the set of countries that can be listed in any order. Paradoxically, despite containing more information, enumerations are easier to remember. The reason for this has been discussed earlier in the context of the minimum information principle: **you should always try to make sure your brain works in the exactly same way at each repetition.** In the case of sets, listing members in varying order at each repetition has a disastrous effect on memory. It is nearly impossible to memorize sets containing

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for dealing with enumerations

III-formulated knowledge – Sets are unacceptable!

Q: What countries belong to the European Union (2002)?

A: Austria, Belgium, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Luxembourg, the Netherlands, Portugal, Spain, Sweden, and the United Kingdom.

Well-formulated knowledge – Converting a set into a meaningful listing

Q: Which country hosted a meeting to consider the creation of a European Community of Defence in 1951?

A: France

Q: Which countries apart from France joined the European Coal and Steel Community in 1952?

A: Germany, Italy and the Benelux

Q: What countries make up the Benelux?

A: Belgium, Luxembourg, and the Netherlands

Q: Whose membership did Charles de Gaulle oppose in the 1960s?

A: that of UK

Q: Which countries joined the EEC along the UK in 1973?

A: Ireland and Denmark

Q: Which country joined the EEC in 1981?

A: Greece

Q: Which countries joined the EEC in 1986?

A: Spain and Portugal

Q: Which countries joined the EU in 1995?

A: Austria, Sweden and Finland

Q: What was the historic course of expansion of the European Union membership?

A: (1) France and (2) Germany, Italy and the Benelux, (3) UK and (4) Ireland and Denmark, (5) Greece, (6) Spain and Portugal and (7) Austria, Sweden and Finland

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as separate items to simplify the learning process. Note also that the sum of information included in this well-formulated approach is far greater than that of the original set. Thus along simplicity, we gained some useful knowledge. All individual items effectively comply with the minimum information principle! You could go further by trying to split the Germany-Italy-Benelux set or using mnemonic techniques to memorize the final seven-member enumeration (i.e. the last of the questions above). However, you should take those steps only if you have any problems with retaining the proposed set in memory.

10) Avoid enumerations

Enumerations are also an example of classic items that are hard to learn. They are still far more acceptable than sets. Avoid enumerations wherever you can. If you cannot avoid them, deal with them using cloze deletions (overlapping cloze deletions if possible). Learning the alphabet can be a good example of an overlapping cloze deletion:

Hard to learn item
Q: What is the sequence of letters in the alphabet? A: abcdefghijklmnopqrstuvwxyz
Easy to learn items
Q: What three letters does the alphabet begin with? A: ABCQ: Fill out the missing letters of the alphabet A E A: B, C, D Q: Fill out the missing letters of the alphabet B F A: C, D, E Q: Fill out the missing letters of the alphabet C G A: D, E, F

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However, once all individual pieces are well remembered, reciting the whole should be a pleasant and speedy action that produces little frustration.

The cloze deletion used above is an overlapping cloze deletion, i.e. the same parts of the enumeration are strengthened in memory using different items (for example, the sequence C-D will be needed to recall the second and the third item). This redundancy does not contradict the minimum information principle because **the extra information is added in extra items**.

You can also deal with enumerations by using grouping like in the case of sets (see the European Union example) but cloze deletions should be simpler and should suffice in most cases.

Learning poems is an example of learning enumerations (all words and sentences have to be uttered in a predefined sequence); however, due to strong semantic connections, the rhyme and the rhythm, it may often be possible to effectively remember poems without using cloze deletion and without the frustration of forgetting small subcomponents again and again. However, once you notice you stumble with your poem, you should dismember it using cloze deletion and thus make sure that the learning is fast, easy, effective and pleasurable

A poem that is hard to remember

Q: The credit belongs ... (Teddy Roosevelt)

A: The credit belongs to the man who's actually in the arena, whose face is marred by dust and sweat; a man who knows the great enthusiasm and the great devotions, who spends himself in a worthy cause, who in the end knows the triumph of high achievement, so that his place shall never be with those cold and timid souls who know neither victory nor defeat

A poem split into easy items

Q: The credit belongs ... (Teddy Roosevelt)

A: to the man who's actually in the arena

Q: The credit belongs to the man who's actually in the arena ...

A: whose face is marred by dust and sweat (a man who knows the great enthusiasm)

Q: whose face is marred by dust and sweat ... (The credit belongs)

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achievement)

Q: who spends himself in a worthy cause ... (The credit belongs)

A: who in the end knows the triumph of high achievement (so that his place shall never be), etc.
etc.

Does it all sound artificial? It does! But you will never know how effective this approach is until you try it by yourself!

11) Combat interference

When you learn about similar things you often confuse them. For example, you may have problems distinguishing between the meanings of the words *historic* and *historical*. This will even be more visible if you memorize lots of numbers, e.g. optimum dosages of drugs in pharmacotherapy. If knowledge of one item makes it harder to remember another item, we have a case of **memory interference**. You can often remember an item for years with straight excellent grades until ... you memorize another item that makes it nearly impossible to remember either! For example, if you learn geography and you memorize that the country located between Venezuela, Suriname and Brazil is Guyana, you are likely to easily recall this fact for years with just a couple of repetitions. However, once you add similar items asking about the location of all these countries, and French Guyana, and Colombia and more, you will suddenly notice strong memory interference and you may experience unexpected forgetting. In simple terms: you will get confused about what is what. Interference is probably the single greatest cause of forgetting in collections of an experienced user of SuperMemo. You can never be sure when it strikes, and the only hermetic procedure against it is to **detect and eliminate**. In other words, in many cases it may be impossible to predict interference at the moment of formulating knowledge. Interference can also occur between remotely related items like Guyana, Guyard and Guyenne, as well as Guyana, kayman and ... aspirin. It may work differently for you and for your colleague. It very hard to predict. Still you should do your best to prevent interference before it takes its toll. This will make your learning process less stressful and mentally bearable. Here are some tips:

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- eliminate interference as soon as you spot it, i.e. before it becomes your obsession (e.g. as soon as you see the word **inept** you think *"I know the meanings of **inept** and **inapt** but I will never know which is which!"*)
- in SuperMemo use **View : Other browsers : Leeches** (*Shift+F3*) to regularly review and **eliminate** most difficult items
- read more: Memory interference

12) Optimize wording

The wording of your items must be optimized to make sure that in minimum time the right bulb in your brain lights up. This will reduce error rates, increase specificity, reduce response time, and help your concentration. **Less optimum item: cloze deletion that is too wordy** Q: Aldus invented desktop publishing in 1985 with PageMaker. Aldus had little competition for years, and so failed to improve. Then Denver-based ... blew past. PageMaker, now owned by Adobe, remains No. 2

Less optimum item: cloze deletion that is too wordy

Q: Aldus invented desktop publishing in 1985 with PageMaker. Aldus had little competition for years, and so failed to improve. Then Denver-based ... blew past. PageMaker, now owned by Adobe, remains No. 2

A: Quark

Better item: fewer words will speed up learning

Q: Aldus invented desktop publishing in 1985 with PageMaker but failed to improve. Then ... blew past (PageMaker remains No. 2)

A: Quark

Or better:

Q: Aldus invented desktop publishing with PageMaker but failed to improve. It was soon outdistanced by ...

A: Quark

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Q: PageMaker lost ground to ...

A: Quark

Note that the loss of information content in this item is inconsequential. During repetition you are only supposed to learn the name: *Quark*. You should not hope that the trailing messages on the ownership of PageMaker and the year of its development will somehow trickle to your memory as a side effect. You should decide if the other pieces of information are important to you and if so, store them in separate items (perhaps reusing the above text, employing cloze deletion again and optimizing the wording in a new way). Otherwise the redundant information will only slow down your learning process!

13) Refer to other memories

Referring to other memories can place your item in a better context, simplify wording, and reduce interference. In the example below, using the words *humble* and *suppliant* helps the student focus on the word *shamelessly* and thus strengthen the correct semantics. Better focus helps eliminating interference. Secondly, the use of the words *humble* and *suppliant* makes it possible to avoid interference of *cringing* with these words themselves. Finally, the proposed wording is shorter and more specific. Naturally, the rules basics-to-details and do not learn what you do not understand require that the words *humble* and *suppliant* be learned beforehand (or at least at the same time)

Item subject to strong interference

Q: derog: adj: shamelessly conscious of one's failings and asking in a begging way

A: cringing

Item that uses interfering memories to amplify the correct meaning

Q: derog: adj: shamelessly humble and suppliant

A: cringing

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trying to paint a picture that would aptly illustrate the question

Item subject to strong interference

Q: What is the name of a soft bed without arms or back?

A: divan

Item that uses interfering memories to amplify the correct meaning

Q: What is the name of a soft bed without arms or back? (like the one at Robert's parents)

A: divan

If you remember exactly what kind of soft bed can be found in Robert's parents' apartment you will save time by not having to dig exactly into the semantics of the definition and/or looking for an appropriate graphic illustration for the piece of furniture in question. Personalized examples are very resistant to interference and can greatly reduce your learning time

15) Rely on emotional states

If you can illustrate your items with examples that are vivid or even shocking, you are likely to enhance retrieval (as long as you do not overuse same tools and fall victim of interference!). Your items may assume bizarre form; however, as long as they are produced for your private consumption, the end justifies the means. Use objects that evoke very specific and strong emotions: love, sex, war, your late relative, object of your infatuation, Linda Tripp, Nelson Mandela, etc. It is well known that emotional states can facilitate recall; however, you should make sure that you are not deprived of the said emotional clues at the moment when you need to retrieve a given memory in a real-life situation

Harder item

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Q: a light and joking conversation (e.g. Mandela and de Klerk in 1992)

A: banter

If you have vivid and positive memories related to the meetings between Nelson Mandela and F.W. de Klerk, you are likely to quickly grasp the meaning of the definition of banter. Without the example you might struggle with interference from words such as *badinage* or even *chat*. There is no risk of irrelevant emotional state in this example as the state helps to define the semantics of the learned concept! A well-thought example can often reduce your learning time several times! I have recorded examples in which an item without an example was forgotten 20 times within one year, while the same item with a subtle interference-busting example was not forgotten even once in ten repetitions spread over five years. This is roughly equivalent to **25-fold saving in time in the period of 20 years!** Such examples are not rare! They are most effectively handled with the all the preceding rules targeted on simplicity and against the interference

16) Context cues simplify wording

You can use categories in [SuperMemo 2000/2002](#), provide different branches of knowledge with a different look (different template), use reference labels (*Title*, *Author*, *Date*, etc.) and clearly label subcategories (e.g. with strings such as *chem* for *chemistry*, *math* for *mathematics*, etc.). This will help you simplify the wording of your items as you will be relieved from the need to specify the context of your question. In the example below, the well-defined prefix *bioch*: saves you a lot of typing and a lot of reading while still making sure you do not confuse the abbreviation GRE with Graduate Record Examination. Note that in the recommended case, you process the item starting from the label *bioch* which puts your brain immediately in the right context. While processing the lesser optimum case, you will waste precious milliseconds on flashing the standard meaning of GRE and ... what is worse ... you will light up the wrong areas of your brain that will now perhaps be prone to interference!

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Context-labeled items increase success rate

Q: bioch: GRE

A: glucocorticoid response element

17) Redundancy does not contradict minimum information principle

Redundancy in simple terms is more information than needed or duplicate information, etc. Redundancy does not have to contradict the minimum information principle and may even be welcome. The problem of redundancy is too wide for this short text. Here are some examples that are only to illustrate that *minimum information principle* cannot be understood as *minimum number of characters or bits in your collections or even items*:

- **passive and active approach:** if you learn a foreign language, e.g. Esperanto, you will often build word pairs such as *phone-telefono*, *language-lingvo*, *hope-esperanto*, etc. These pairs require active recall of the foreign word. Active recall does not, however, guarantee passive recognition and you may fail with *telefono-phone*, *lingvo-language*, or *esperanto-hope*. Adding new elements with swapped questions and answers may in some cases be redundant but it does not contradict the minimum information principle! Your items are still as simple as possible. You just get more of them! In [SuperMemo 2000/2002](#), you can quickly generate swapped word-pair items with **Duplicate** (*Ctrl+Alt+D*) and **Swap** (*Ctrl+Shift+S*)
- **reasoning cues:** you will often want to boost your reasoning ability by asking about a solution to the problem. Instead of just *memorizing* the answer you would like to quickly follow the reasoning steps (e.g. solve a simple mathematical equation) and *generate* the answer. In such a case, providing the hint on the reasoning steps in the answer will only serve helping you always follow the right path at repetitions
- **derivation steps:** in more complex problems to solve, memorizing individual derivation steps is always highly recommended (e.g. solving complex mathematical problems). It is not cramming! It is making sure that the brain can always follow the fastest path while solving the problem. For more on boosting creativity and intelligence read: *Roots of genius and creativity*, as well as more specific: *Derivation, reasoning and intelligence*
- **multiple semantic representation:** very often the same knowledge can be represented and viewed from different angles. Memorizing different representations of the same fact or rule is recommended in cases

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providing just one of the equivalent choices. For example, if you learn a language, it rarely make sense to learn all synonyms that meet a definition of a concept. It is more adequate to consider a single synonym as the sufficient answer (e.g. *a mark made by ink spilt on sth = blot/blob/blotch*)

● more

18) Provide sources

Except for well-tested and proven knowledge (such as $2+2=4$), it is highly recommended that you include sources from which you have gathered your knowledge. In real-life situation you will often be confronted with challenges to your knowledge. Sources can come to your rescue. You will also find that facts and figures differ depending on the source. You can really be surprised how frivolously reputable information agencies publish figures that are drastically different from other equally reputable sources. Without SuperMemo, those discrepancies are often difficult to notice: before you encounter the new fact, the old one is often long forgotten. With sources provided, you will be able to make more educated choices on which pieces of information are more reliable. Adding reliability labels may also be helpful (e.g. *Watch out!*, *Other sources differ!*, etc.). Sources should accompany your items but should not be part of the learned knowledge (unless it is critical for you to be able to recall the source whenever asked).

19) Provide date stamping

Knowledge can be relatively stable (basic math, anatomy, taxonomy, physical geography, etc.) and highly volatile (economic indicators, high-tech knowledge, personal statistics, etc.). It is important that you provide your items with time stamping or other tags indicating the degree of obsolescence. In case of statistical figures, you might stamp them with the year they have been collected. When learning software applications, it is enough you stamp the item with the software version. Once you have newer figures you can update your items. Unfortunately, in most cases you will have to re-memorize knowledge that became outdated. Date stamping is useful in editing and verifying your knowledge; however, you will rarely want to memorize stamping itself. If you would like to remember the changes of a given figure in time (e.g. GNP figures over a number of years), the date stamping becomes the learned knowledge itself.

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your knowledge slots in. This will also affect the speed of learning (e.g. see: learn basics first). There are many stages at which prioritizing will take place; only few are relevant to knowledge representation, but all are important:

- 1 Prioritizing sources** – there will always be a number of sources of your knowledge. If you are still at student years: these will most likely be books and notes pertaining to different subjects. Otherwise you will probably rely more on journals, Internet, TV, newspapers, encyclopedias, dictionaries, etc. It is always worth being aware what is the optimum proportion of time devoted to those varied sources. As you progress with learning, you will quickly develop a good sense of which learning slots bring better results and which might be extended at the cost of others
- 2 Extracting knowledge** – unless you are about to pass an important exam, it nearly never makes sense to memorize whole books or whole articles. You will need to extract those parts that are most likely to impact the quality of your knowledge. You can do it by (1) marking paragraphs in a book or journal, (2) pasting relevant web pages to SuperMemo, (3) pasting relevant passages to SuperMemo, (4) typing facts and figures directly to SuperMemo notes, etc. You will need some experience before you can accurately measure how much knowledge you can indeed transfer to your brain and what degree of detail you can feasibly master. Your best way to prioritize the flow of knowledge into your memory is to use incremental reading tools
- 3 Transferring knowledge to SuperMemo** – you may try to stick with the 20 rules of formulating knowledge at the moment of introducing your material to SuperMemo. However, you can also literally transfer your notes or import whole files and later use the mechanisms provided by SuperMemo to determine the order of processing the imported material. Probably the best criterion for choosing between formulating or just importing is the time needed for accurately formulating the item or items. If formulation requires more knowledge, more time, comparing with other sources, etc. you can just import. Otherwise, if you believe that formulating an accurate item is a matter of seconds, formulate it
- 4 Formulating items** – make sure that explanatory or optional components of the answer are placed in the parentheses so that your attention is focused on the most important part of the item. The parts in the parentheses can be read after the repetition to strengthen the memory in its context
- 5 Using forgetting index** – you can use the forgetting index to prioritize pending items. The sequence of repetitions will naturally be determined by SuperMemo; however, you can request higher retention level for items that are more important and lower retention level for items of lower priority
- 6 Learning** – the process of prioritizing does not end with the onset of repetitions. Here are the tools you can use to continue setting your priorities while the learning process is under way:

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- **Reschedule** (*Ctrl+J*) – manually schedule the date of the next repetition
- **Execute repetition** (*Ctrl+Shift+R*) – manually execute a repetition even before the repetition's due date (e.g. when reviewing particularly important material)
- **Forget** (*Ctrl+R*) – remove the current item from the learning process and place it at the end of the pending queue
- **Dismiss** (*Ctrl+D*) – ignore the current item in the learning process altogether
- **Delete** (*Ctrl+Shift+Del*) – remove the current item from your collection
- Change the forgetting index of memorized items or change the ordinal of pending items (*Ctrl+Shift+P*)

Summary

Here again are the twenty rules of formulating knowledge. You will notice that the first 16 rules revolve around making memories simple! Some of the rules strongly overlap. For example: do not learn if you do not understand is a form of applying the minimum information principle which again is a way of making things simple:

- 1 **Do not learn if you do not understand**
- 2 **Learn before you memorize** – build the picture of the whole before you dismember it into simple items in SuperMemo. If the whole shows holes, review it again!
- 3 **Build upon the basics** – never jump both feet into a complex manual because you may never see the end. Well remembered basics will help the remaining knowledge easily fit in
- 4 **Stick to the minimum information principle** – if you continue forgetting an item, try to make it as simple as possible. If it does not help, see the remaining rules (cloze deletion, graphics, mnemonic techniques, converting sets into enumerations, etc.)
- 5 **Cloze deletion is easy and effective** – completing a deleted word or phrase is not only an effective way of learning. Most of all, it greatly speeds up formulating knowledge and is highly recommended for beginners

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- 8 **Graphic deletion is as good as cloze deletion** – obstructing parts of a picture is great for learning anatomy, geography and more
- 9 **Avoid sets** – larger sets are virtually un-memorizable unless you convert them into enumerations!
- 10 **Avoid enumerations** – enumerations are also hard to remember but can be dealt with using cloze deletion
- 11 **Combat interference** – even the simplest items can be completely intractable if they are similar to other items. Use examples, context cues, vivid illustrations, refer to emotions, and to your personal life
- 12 **Optimize wording** – like you reduce mathematical equations, you can reduce complex sentences into smart, compact and enjoyable maxims
- 13 **Refer to other memories** – building memories on other memories generates a coherent and hermetic structure that forgetting is less likely to affect. Build upon the basics and use planned redundancy to fill in the gaps
- 14 **Personalize and provide examples** – personalization might be the most effective way of building upon other memories. Your personal life is a gold mine of facts and events to refer to. As long as you build a collection for yourself, use personalization richly to build upon well established memories
- 15 **Rely on emotional states** – emotions are related to memories. If you learn a fact in the state of sadness, you are more likely to recall it if when you are sad. Some memories can induce emotions and help you employ this property of the brain in remembering
- 16 **Context cues simplify wording** – providing context is a way of simplifying memories, building upon earlier knowledge and avoiding interference
- 17 **Redundancy does not contradict minimum information principle** – some forms of redundancy are welcome. There is little harm in memorizing the same fact as viewed from different angles. Passive and active approach is particularly practicable in learning word-pairs. Memorizing derivation steps in problem solving is a way towards boosting your intellectual powers!
- 18 **Provide sources** – sources help you manage the learning process, updating your knowledge, judging its reliability, or importance
- 19 **Provide date stamping** – time stamping is useful for volatile knowledge that changes in time
- 20 **Prioritize** – effective learning is all about prioritizing. In *incremental reading* you can start from badly formulated knowledge and improve its shape as you proceed with learning (in proportion to the cost of inappropriate formulation). If need be, you can review pieces of knowledge again, split it into parts,

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